

Samuel Okafor

Embedded Systems Engineer

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SUMMARY

Embedded systems engineer with 6 years of experience developing firmware for consumer IoT and industrial devices. Comfortable working across the stack from low-level driver code to real-time operating systems and hardware bring-up.

CORE SKILLS

Languages	C, C++, Rust, Python (tooling)
Platforms	ARM Cortex-M, FreeRTOS, Zephyr RTOS
Protocols	I2C, SPI, UART, BLE, MQTT
Tools	JTAG/SWD debugging, oscilloscopes, logic analyzers, Git

EXPERIENCE

Embedded Systems Engineer

Ironforge Devices — Phoenix, AZ

Jan 2022 - Present

- Develop firmware in C and Rust for a line of industrial IoT sensors deployed across 3,000+ field installations.
- Led the bring-up of a new sensor board from schematic review through production firmware, shipping on schedule for a major customer contract.
- Reduced average device power consumption by 38% through low-power mode optimization, extending battery life from 8 months to over a year.
- Built an OTA firmware-update system with rollback protection, reducing failed-update-related field visits by 90%.

Firmware Engineer

Solara Home Electronics — Tempe, AZ

Jun 2019 - Dec 2021

- Wrote and maintained BLE communication firmware for a smart-lock product line shipping 200,000+ units annually.
- Debugged and resolved an intermittent connectivity issue affecting 4% of devices in the field, tracing it to a race condition in the BLE stack.
- Built the automated hardware-in-the-loop test rig used to validate firmware builds before release, cutting manual QA time by 70%.

EDUCATION

Arizona State University

B.S. in Electrical Engineering

2015 - 2019